

# DAEN 300 Data Engineering Coding Experience I

Fall 2026

## Course Information

**Time (Location)** Fri 12:40–3:30 PM *0 Lecture Hours. 3 Lab Hours.* (ZACH 344)

**Prerequisites** Grade of C or better in ECEN 360; concurrent enrollment in DAEN 321 and DAEN 331; junior or senior.

**Instructor** Alexander Abuabara <mailto:abuabara@tamu.edu>

**Course office hours (location)** Wed 1-3 PM or by appointment (ETB 4013)

## Description

Application of computational tools to model and solve data engineering problems primarily involving machine learning and optimization techniques.

## Outcomes

1. Use Python and Jupyter to solve data engineering problems.
2. Clean, transform, and analyze data using NumPy and pandas.
3. Apply basic algorithms, including recursion, searching, and sorting.
4. Build and evaluate basic machine learning models.
5. Use optimization techniques to improve computational solutions.
6. Create and explain an end-to-end data analysis and machine learning workflow.

## Recommended Resources

- Python Data Science Handbook (PDSH) <https://jakevdp.github.io/PythonDataScienceHandbook/>.
- The Recursive Book of Recursion (RBR) <https://inventwithpython.com/recursion/>.

## Grading

| Assessment                 | Weight | Description                                              |
|----------------------------|--------|----------------------------------------------------------|
| Weekly Assignments (12–13) | 89%    | Weekly activities and worksheets; each weighted equally. |
| In-Class Participation     | 11%    | In-class quizzes, discussions, participation             |

Grades are based on total points earned: A ≥ 90%, B ≥ 80%, C ≥ 70%, D ≥ 60%, F < 60%.

## Schedule (tentative)

| Week | Topic                                                    | Reading              | Lab/Coding                                 |
|------|----------------------------------------------------------|----------------------|--------------------------------------------|
| 1    | Python and Jupyter: Functions, debugging, profiling      | <i>PDSH 1, RBR 1</i> | Load and explore a dataset in Jupyter      |
| 2    | NumPy: Arrays, indexing, vectorization, aggregation      | <i>PDSH 2</i>        | Process numerical data                     |
| 3    | (pandas) DataFrames, filtering, missing data, joins      | <i>PDSH 3</i>        | Clean and combine datasets                 |
| 4    | Data transformation: grouping, pivot tables, strings     | <i>PDSH 3</i>        | Summarize and transform data               |
| 5    | Recursive + iterative solutions, base cases, call stacks | <i>RBR 1+2</i>       | Compare recursive and iterative programs   |
| 6    | Algorithms: binary search, merge sort, quick sort        | <i>RBR 3+5</i>       | Implement searching and sorting algorithms |
| 7    | Matplotlib: Visualization and exploratory analysis       | <i>PDSH 4</i>        | Explore and visualize a dataset            |
| 8    | Scikit-Learn: Machine learning basics                    | <i>PDSH 5</i>        | Build a basic machine learning pipeline    |
| 9    | Linear regression                                        | <i>PDSH 5</i>        | Build, evaluate a regression model         |
| 10   | Classification, model evaluation                         | <i>PDSH 5</i>        | Train and compare classification models    |
| 11   | Decision trees, random forests, tree traversal           | <i>PDSH 5, RBR 4</i> | Build tree-based models                    |
| 12   | Clustering, PCA, model selection                         | <i>PDSH 5</i>        | Apply clustering, dimensionality reduction |
| 13   | Optimization: backtracking, dynamic programming          | <i>RBR 6-7-8</i>     | Solve, improve an optimization problem     |
| 14   | Mini-project: Data prep → ML → optimization              | –                    | An end-to-end data engineering solution    |

## Policies

- Be nice. Be honest. Don't cheat. Adhere to all University Policies.
- Course material is cumulative and requires consistent effort. All assignments must be completed carefully and on time. Presentation quality (organization, clarity, tables, graphs, and discussion) is part of grading.
- Excused absences apply only to attendance and in-class activities and do not extend assignment deadlines. Alternative arrangements must be made prior to the deadline.
- Regrade applies to the entire submission. To ensure fairness and timely feedback, no changes after one week.
- All assignments must be submitted through Canvas. If technical issues prevent submission, send your assignment from your @tamu.edu email to the instructor's @tamu.edu email as a backup. The subject line must include the course number and assignment name. Late work is not accepted and will receive a grade of zero.
- Make-up assignments may be given orally upon the student's return from a documented, verifiable absence, on a date set by the professor.

- Audio or video recording of lectures or any portion of the course is prohibited without written instructor permission.
- Headphones and non-class-related browsing are not permitted during class.
- Use of others' ideas is encouraged as part of scholarly work, but all sources must be properly cited. Proper attribution is required for all coursework and projects.